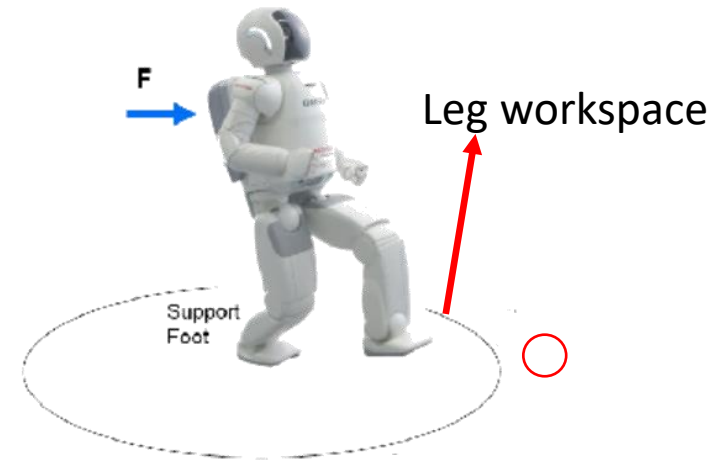


Balance Metrics

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Balance recovery strategies

- ① move CoP under the COM (inside support polygon)
 - ② create centroidal moments by wind-milling or squatting (hip strategy)
 - ③ Take a step (problem of finding the position to step with a point-foot robot to recover from a push)
- Where/when Take a step?
 - **capture point** = point on the ground where the robot can step to arrive to a complete stop.
 - **capture region** = collection of all capture points
 - CP can be computed in closed form only for simple models (eg. LIP)
 - if CP is INSIDE the support polygon the robot can recover WITHOUT taking a step
 - if CP is OUTSIDE the support polygon the robot needs to step to be able to "RECAPTURE" it and stop
 - if CP is OUTSIDE the workspace of a leg the robot will fall (unless it takes more steps)
 - **capture state** = robot state where the kinetic energy of a biped is zero and can remain zero (i.e. COM above ZMP)



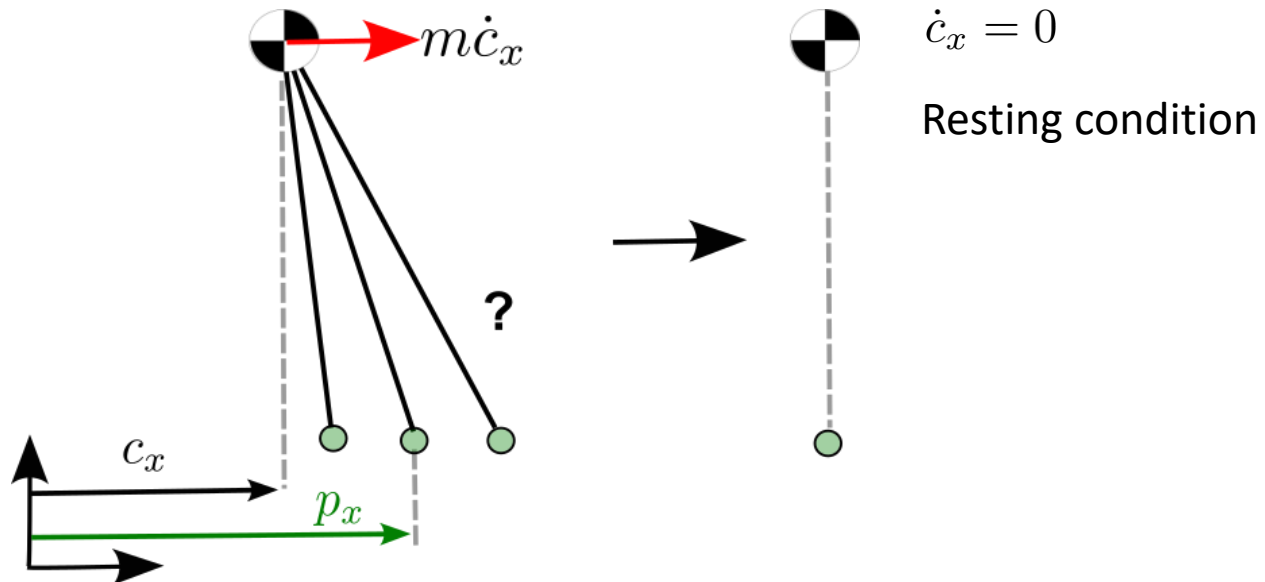
Capture point for LIP

[Pratt and Goswami, 2006]

- Recalling the LIP dynamics

$$\begin{aligned} (1) \quad & \begin{bmatrix} c_x(t) \\ \dot{c}_x(t) \end{bmatrix} = \begin{bmatrix} \cosh(\omega t) & \frac{1}{\omega} \sinh(\omega t) \\ \omega \sinh(\omega t) & \cosh(\omega t) \end{bmatrix} \begin{bmatrix} c_{x0} \\ \dot{c}_{x0} \end{bmatrix} + \begin{bmatrix} 1 - \cosh(\omega t) \\ -\omega \sinh(\omega t) \end{bmatrix} z_x \\ (2) \quad & \end{aligned}$$

- Given the initial position c_{x0} , and velocity $\dot{c}_{x0}$ (e.g. after a push) where should put the **foot** such that $c_x \rightarrow p_x$, and remains at a stop at the end of the motion ($\dot{c}_x = 0$)?
- Assume the robot places the foot $p_x = z_x$ and consider this input constant



Capture point for LIP

- Such a simple problem can be solved **analytically**, without resorting to optimization

$$(1) \quad \lim_{t \rightarrow \infty} c_x(t) = \cancel{p_x} = c_{x0} \cosh \omega t + \frac{\dot{c}_{x0}}{\omega} \sinh \omega t + \cancel{p_x} - p_x \cosh \omega t$$

$$c_{x0} \cosh \omega t + \frac{\dot{c}_{x0}}{\omega} \sinh \omega t = p_x \cosh \omega t \quad \text{Collect cosh()}$$

$$c_{x0} \cosh \omega t + \frac{\dot{c}_{x0}}{\omega} \underbrace{\frac{\sinh \omega t}{\cosh \omega t}}_{\tanh \omega t} = p_x \quad \tanh(\omega t) \rightarrow 1 \quad \text{as } t \rightarrow \infty$$

$$(3) \quad \boxed{p_x = c_{x0} + \frac{\dot{c}_{x0}}{\omega} \triangleq \varepsilon_x} \quad \text{Capture point (CP)/divergent component of motion (DCM)}$$

- Let's verify that $\dot{c}_x = 0$ for $t \rightarrow \infty$ insert p_x into (2)

$$\begin{aligned} \dot{c}_x &= \omega \sinh \omega t c_{x0} + \cos \omega t \dot{c}_{x0} - \omega \sin \omega t \left(c_{x0} + \frac{\dot{c}_{x0}}{\omega} \right) \\ &= \dot{c}_{x0} (\cosh \omega t - \sinh \omega t) \rightarrow_{t \rightarrow \infty} 0 \quad \text{Q.E.D.} \end{aligned}$$

$$\frac{e^{\omega t} + e^{-\omega t}}{2} - \frac{e^{\omega t} - e^{-\omega t}}{2} = e^{-\omega t}$$

Capture point dynamics

(3) $\varepsilon_x = c_x + \frac{1}{\omega} \dot{c}_x \Rightarrow \dot{c}_x = -\omega(c_x - \varepsilon)$ negative eigenvalue $\Rightarrow$ Has **stable** 1 order dynamics

- Differentiate (3)

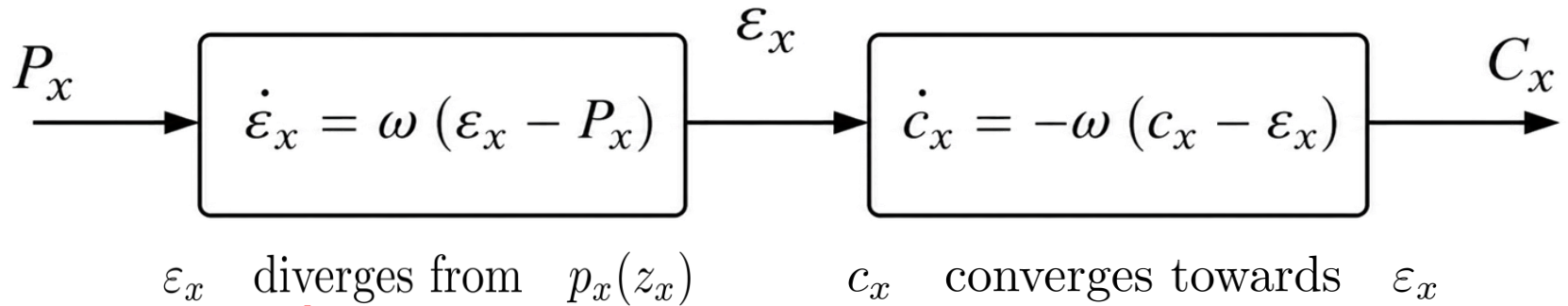
$$\dot{\varepsilon}_x = \dot{c}_x + \frac{1}{\omega} \ddot{c}_x \quad \oplus \quad \ddot{c}_x = \omega^2 (c_x - p_x) \quad (\text{LIP dynamics})$$

$$\begin{aligned} \dot{\varepsilon}_x &= \underbrace{\dot{c}_x + \omega c_x}_{\omega \varepsilon} - \omega p_x \\ &= \omega \varepsilon - \omega p_x = \omega (\varepsilon - p_x) \end{aligned}$$

(4) $\dot{\varepsilon} = \omega(\varepsilon - p_x)$ positive eigenvalue $\Rightarrow$ **unstable** 1 order dynamics

$\Rightarrow$ CP is **repelled** by ZMP

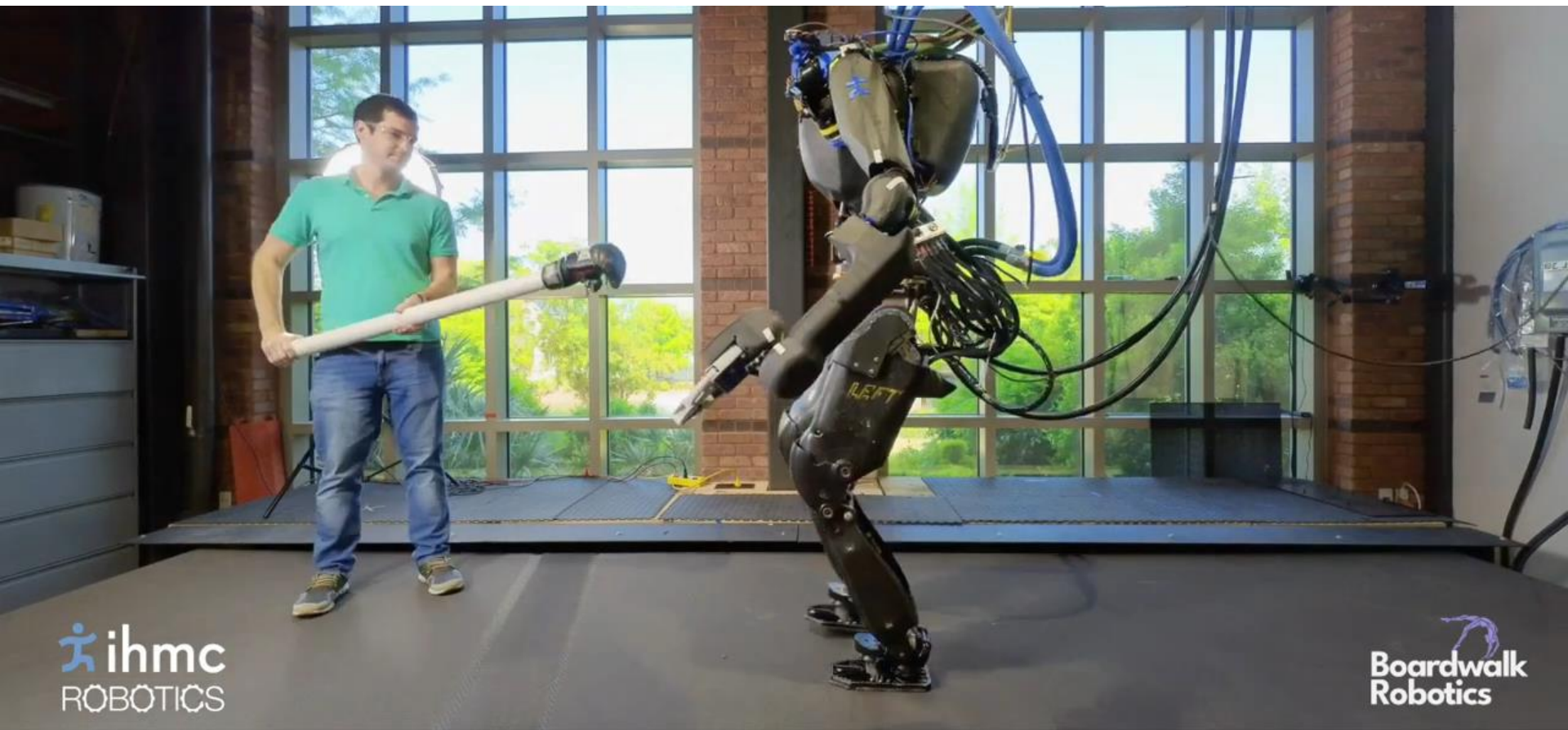
Capture Point based Stepping strategy



$$\begin{aligned} \epsilon(t) &= \epsilon_0 e^{\omega t} + (1 - e^{\omega t}) p_x \\ &= p_x + \underbrace{e^{\omega t}}_{\rightarrow \infty} (\epsilon_0 - p_x) \rightarrow \text{diverges along the line } (\epsilon_x - p_x) \end{aligned}$$

- **STRATEGY:** if the robot encloses the CP in its support polygon AND decide to maintain the ZMP on the CP
 - We nullify the effect of the divergent dynamics
 - the CoM will be attracted by CP
 - it will exist a trajectory leading the CoM to a CAPTURE STATE!

Capture point based push recovery



The capture region can be extended to multiple steps, each with knowledge of the kinematic reachability constraints on where the robot can step.

Capture point applied to landing control

Landing Controller



Naive Approach



[Roscia et. al Reactive Landing Controller for Quadruped Robots, 2023]

Height: 1.00 m
Horizontal height: 1.2 m/s, -90°

